

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2025

SA7 - Investment and Finance

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q. 1) You are the Investment actuary of ABC Life Insurance Co., a large Indian life insurer writing a mix of with-profit (participating) policies, non-participating guaranteed plans, unit-linked insurance plans (ULIPs), and annuity contracts. The company's investment portfolio is currently allocated chiefly to government bonds and high-grade corporate bonds, with a smaller equity portfolio supporting the with-profit fund. The economic environment is challenging: inflation has been persistently high and interest rates have been rising. The Insurance Regulatory and Development Authority of India (IRDAI) is introducing a new Risk-Based Capital (RBC) framework next year, which is expected to significantly increase the capital required for riskier assets. ABC Life's latest solvency ratio under the current rules is adequate but only marginally above the required minimum. Preliminary RBC calculations indicate a potential capital shortfall if the investment strategy remains unchanged.

The Chief Financial Officer (CFO) is concerned about the company's capital position and is exploring ways to strengthen it. One option under consideration is raising additional capital – either by issuing new equity or by raising subordinated debt – to meet the RBC requirements and support future growth. Meanwhile, the Chief Investment Officer (CIO) argues that the company can improve its solvency and profitability by adjusting its investment mix. The CIO proposes allocating a greater portion of the portfolio to higher-yielding assets such as lower-rated corporate bonds, infrastructure debt funds, real estate, and equity, within the limits allowed by IRDAI regulations (for example, the limits on 'Other Investments'). The CIO believes this could boost investment returns and thereby improve surplus over time. In contrast, the Chief Risk Officer and the CFO worry that increasing asset risk could jeopardize the company's ability to meet the guaranteed liabilities, especially under the upcoming RBC regime which assigns high capital charges to equities and other high-risk assets. They suggest that a better approach is to de-risk the portfolio: increase the matching of assets to liabilities (for example, investing more in long-term government bonds or using derivatives to hedge guarantees), in order to stabilize solvency.

The Board has asked you, as the investment actuary, to analyse these issues and advise on a suitable course of action. Answer the following questions based on the scenario above:

- i) **Product Liability Characteristics:** Identify the *four main product groups* written by ABC Life mentioned above, and briefly describe one key investment characteristic or requirement for each. (5)
- ii) **Impact of Inflation and Interest Rates:** Explain how the current environment of high inflation and rising interest rates can affect ABC Life's assets and liabilities. What implications do these economic factors have for the insurer's investment strategy and ability to meet policyholder expectations? (5)
- iii) **Capital Raising Options:** The CFO is considering raising additional capital to improve ABC Life's solvency under the RBC framework. Compare the merits of raising new equity capital versus issuing subordinated debt from the insurer's perspective. In your answer, discuss how each option might impact the company's balance sheet, cost of capital, control/dilution, and financial flexibility. (8)
- iv) **Higher-Yield Investment Strategy:** The CIO proposes to invest a greater portion of the portfolio in higher-yielding instruments (such as lower-rated corporate bonds, infrastructure assets, and equities) to boost returns. Discuss the advantages and risks of this strategy for ABC Life. In your answer, consider factors such as potential investment income vs. asset quality, liquidity, regulatory investment limits, and the effect of the new RBC capital charges. (10)
- v) **De-risking and Asset–Liability Matching:** An alternative approach is to de-risk the portfolio by increasing asset–liability matching and hedging. Discuss the potential benefits and drawbacks of shifting towards a more conservative investment strategy (for example, holding more long-term government bonds or using derivatives to hedge guarantees) in the context of ABC Life's situation. Consider the impact on solvency volatility, guarantee security, long-term returns, and any practical or regulatory constraints. (7)
- vi) **Recommendation:** Given the trade-offs identified above, recommend an overall investment and financing strategy for ABC Life. Your answer should outline a coherent plan that balances the need

to improve returns and solvency with prudent risk control. Justify how your recommended strategy would meet the company's obligations to policyholders (including guarantees) while satisfying regulatory requirements (such as the RBC constraints). (15)

[50 Marks]

Q.2) You are the chief investment officer (CIO) at ABC Asset Management Company (AMC), managing a range of investment funds catering to both retail and institutional investors in India. The macroeconomic environment is experiencing significant stress:

- Global trade tensions and a shift towards protectionist trade policies in the U.S. are impacting capital flows.
- Domestic institutional investors (DIIs) are facing challenges due to demographic shifts, particularly an aging population, affecting their capital generation abilities.
- The Indian stock market is highly volatile, particularly in the Financial Services and Information Technology (IT) sectors, while regulatory changes are influencing portfolio strategies.

ABC AMC holds a USD-denominated bond portfolio worth \$250 million and you are tasked with assessing the risks and recommending a strategy to mitigate the impacts of these macroeconomic and regulatory factors.

Today's spot rate is ₹ 85 = US \$ 1.

i) Global Trade Tensions and the USD/INR Exchange Rate

- Explain how global trade tensions and the U.S.'s protectionist stance may impact the USD/INR exchange rate and capital flows into India. How might these factors affect FII inflows into the Indian equity market? (10)

ii) Impact of INR Depreciation on the Bond Portfolio

- Estimate the impact of a 6% appreciation or 6 % depreciation of INR on the value of ABC AMC's USD-denominated bond portfolio worth \$250 million in INR term.
- What actions can the AMC take to hedge this currency risk? (7)

iii) Tactical Rebalancing in Response to Sector-Specific Risk

- Given the current portfolio allocation (40% Financial Services, 20% IT, 15% Consumer Goods, 15% Healthcare, 10% Energy), if the Financial Services sector is expected to decline by 12%, estimate the overall portfolio loss and recommend a tactical rebalancing strategy to mitigate this risk. (8)

[25 Marks]

Regulatory Considerations:

- SEBI's Liquidity Management Norms: Ensure that the AMC adheres to SEBI's regulations on liquidity, especially when rebalancing the portfolio.
- RBI's Foreign Investment Guidelines: Monitor how changes in FDI and FPI regulations may affect capital flows and market performance.

Q.3) You are an investment consultant advising the Trustees of a large, mature Defined Benefit (DB) Exempt Provident Fund Trust in India, regulated implicitly under EPF & MP Act guidelines and Trustee responsibilities. The scheme is closed to new entrants and future accrual. Its funding level on an actuarial basis is 105%, but the Trustees are concerned about managing assets against liabilities to ensure benefit security, especially given the prevailing interest rate environment in India and the increasing pensioner proportion. The scheme currently holds 60% in Indian equities (mix of large and mid-cap) and 40% in a mix of Indian Government Securities (G-Secs) and high-rated PSU/Corporate

Bonds. The sponsoring employer's covenant is considered reasonably strong but not guaranteed indefinitely.

Prepare a report to the Trustees advising on a revised investment strategy. Your report should:

- i) Analyse the key objectives and constraints for the scheme's investment strategy at its current stage of maturity, considering the Indian regulatory environment for such trusts. Critically evaluate the relevance of the sponsor covenant strength in setting the strategy. (6)
- ii) Explain the principles of Liability-Driven Investment (LDI). Discuss the suitability of different LDI implementation approaches (e.g., using physical Indian bonds like G-Secs/SDLs vs. synthetic replication using derivatives like Overnight Index Swaps referencing MIBOR/Modified-MIBOR or INR Interest Rate Swaps) for this scheme, including the operational considerations of using derivatives within the Indian context. (8)
- iii) Recommend a revised Strategic Asset Allocation (SAA), justifying your choices for both the matching and growth portfolios, suitable for the Indian market. Discuss the rationale for including specific Indian asset classes beyond traditional G-Secs and equities (e.g., AAA/AA corporate bonds, Infrastructure Investment Trusts (InvITs), potentially Alternative Investment Funds (AIFs) within regulatory limits) and the risks they introduce. (7)
- iv) Outline the key governance considerations and processes the Trustees should have in place to effectively oversee the implementation and ongoing management of the revised strategy, including monitoring LDI effectiveness and manager performance within the Indian governance framework for trusts. (4)

[25 Marks]
